Yiyu Zhu

Full-Stack Software Engineer

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Builds Node.js and Python applications with persistent state, failure recovery and reproducible tests. Supports AI application integration at China Telecom.

Stack: JavaScript / Node.js 24 · Python · SQL / SQLite · HTTP APIs · pandas / openpyxl · Streamlit · pytest · GitHub Actions

RunLedger Durable Webhook Delivery

Independent project · Node.js 24 · SQLite · Browser console · Nov 2026 · Source and verification

Delivers AI job completion events and exposes retries, receiver outages and dead-letter recovery in an operator console.

- Commits events and queue records together. Canonical payloads deduplicate by idempotency key; conflicts return HTTP 409. Expiring leases fence stale worker completions.
- Signs HTTP bodies with HMAC. A persistent inbox deduplicates redelivery after a lost acknowledgment; replay preserves delivery IDs and earlier attempt generations.
- Adds persistent dispatch pause/resume and atomic batch replay with expected-generation checks, idempotent operation receipts and an audit view. Verifies competing operator processes and lost-response recovery.
- Uses indexed lookup across the active ledger and up to 20 signed archives. A bounded HMAC-verification cache rechecks file identities around every query; signed activation installs a compact candidate and retains a rollback source.
- Ninety-nine Node.js scenarios cover recovery, archives, activation/concurrency, cache freshness, query plans and HTTP contracts. A local 20k-active/10k-archive, four-archive lookup measured 97.99 ms cold and 29.63 ms warm (3.3x). Linux/Windows and browser CI pass.

Selected Projects

OrderCheck Order Data Quality Workbench Validates money with Decimal, isolates conflicting order groups and exports a seven-sheet audit workbook. Publishes ZIP reports exclusively and replaces KPI JSON after staging and sync. Seventy-eight tests cover rules, provenance, UI/CLI behavior, invalid KPI input and export failures.

PriceWatch Historical Monitoring Commits SQLite snapshots and recoverable export intents together. Incremental CSV recovery hashes and writes one saved chunk at a time; HTML and alert JSON use durable receipts. Backlog age supports health probes; 78 tests cover recovery and concurrent exporters.

LeadReview and API Contract Lab Separates AI extraction, deterministic scoring and human review. Generated n8n code is tested without module globals. HTTP/JSON/SSE regression tooling passes 18 self-tests; lead rules have 17 explicit assertions.

Experience

China Telecom Innovation Product Operations Center · Jun 2026-Present

- Support AI application integration: reproduce recognition, configuration and event-delivery failures; adjust conditions and retest to isolate the cause.
- Record expected and actual behavior, reproduction steps and retest results; maintain configuration checks for project support.

Education and Languages

City University of Hong Kong · MSc, Electronic Information Engineering · Sep 2025-Jun 2026

Sun Yat-sen University · BEng, Aerospace Engineering · Sep 2021-Jun 2025

Native Mandarin · English CET-6 505 · English-medium postgraduate study